Diego Emiliano Navarro Arellano

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Education

Instituto Politécnico Nacional (ESCOM) — Mexico City, Mexico
B.S. in Computer Systems Engineering | GPA: 9.15/10.0 — *Expected January 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Theory of Computation, Computer Architecture, Computer Networks, Systems Analysis and Design.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL
Frameworks: React, Node.js, Express, Zustand, Hono
Databases: PostgreSQL, MySQL, Supabase, Redis
Tools: Git, Linux, Render, Tailwind, Bootstrap, Pygame, Socket.io, BullMQ
Spoken Languages: Spanish (Native), English (B2 - Conversational)

Projects

Tlamatini (Social Network & Travel Planner) | *React, Node.js, Supabase* — December 2025

- Developed a full-stack social web application to create modular travel itineraries and share location-based posts.
- Implemented a real-time messaging system and content moderation features to enhance user safety.

NFA Chess Simulator | *Python, Pygame* — April 2025

- Engineered a 4x4 chess simulation by modeling board mechanics as a Non-Deterministic Finite Automaton (NFA).
- Implemented a DFS algorithm and Manhattan distance heuristic to calculate optimal move sequences.
- Developed a dynamic graphical interface in Pygame to render real-time layered NFA state transitions.

Logo-like Language Compiler | *Java* — January 2026

- Constructed a custom compiler interpreting commands to generate geometric figures and graphics.
- Implemented lexical and syntax analysis using a recursive descent parser to render visual outputs based on user-defined code.

Parkify (Access Control System) | *React, Node.js, Supabase* — January 2026

- Built a parking management platform with CRUD functionalities for access control and role-based permissions.
- Designed an automated reporting module to track real-time parking availability and historical usage data.

Extracurricular Experience & Achievements

Hackathon 2nd Place Winner – Coppel Track — April 2026
Talent Land: "Cancha Justa para el Mundial 2026" — *Mexico City, Mexico*

- Secured second place out of multiple teams by developing an innovative software solution, earning an electronic monetary prize from Fundación Coppel.

Competitive Programmer — September 2023 – Present
ESCOM Algorithmics Club — *Mexico City, Mexico*

- Solved complex algorithmic and data structure challenges on platforms like Codeforces to optimize execution time and memory limits.

Speaker / Researcher — May 2025
3rd Science Divulgation Journey (IPN) — *Mexico City, Mexico*

- Presented the accepted paper "The Use of Artificial Intelligence for the Discovery of Conjectures and Making Mathematical Advances".
- Showcased research findings at a high-level academic event hosted by ESCOM.